MAIS PAC

IPhO 01.01: Working as a Physicist: Approximation, Symmetry and Units · Lecture + Problems

Working as a Physicist: Approximation, Symmetry and Units

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

7 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. What the first page of every problem is really asking

Physics olympiads do not test whether you can substitute numbers faster than a calculator. They test whether you can turn a messy description into a model. A model is a short list of decisions: which objects have mass, which forces matter, which effects you throw away and why. Once the model is written down, the mathematics is usually ordinary.

There is a reliable working order:

1. **Sketch** the geometry and label the quantities with symbols before any numbers appear.
2. **Name the regime.** Is it fast or slow, dense or dilute, quantum or classical, relativistic or Newtonian?
3. **Look for symmetry** before writing equations. A symmetry can delete an unknown outright.

4. **Check dimensions** as soon as you have an expression.
5. **Take a limit.** If your formula fails in an obvious limit, it is wrong.
6. **Estimate the answer** before computing it, so a factor-of-ten mistake announces itself.

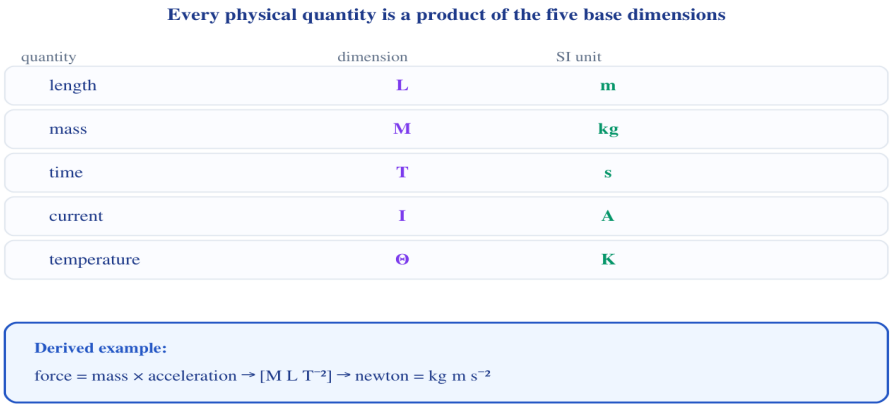


Figure 1. If two sides of an equation have different dimensions, the equation is wrong. Dimensional checking is free, instant and exact.

2. Units and dimensions: the free check

Every mechanical quantity is a product of powers of mass [M], length [L] and time [T] (add current [I] and temperature [] for electricity and heat). A force has dimensions [M,L,T^(-2)]; energy has [M,L²,T^(-2)].

Two rules do all the work:

- **Both sides of an equation carry the same dimensions.** Adding a length to a time is meaningless, no matter how nice the algebra looks.
- **Only dimensionless groups can be arguments of a function** such as sin, exp or ln. If you find yourself writing exp (L), the expression is wrong.

Dimensional analysis cannot give you the pure number in front. From T ∝ √(L/g) you learn the exponent (1)/(2) but not 2π. That is still a great deal: it tells you how the period changes when you change the length, and it rules out whole families of wrong formulas.

Interactive experiment: Dimensional analysis as a searchlight — open the online handout to use it.

3. Scaling laws: the whole physics in one exponent

Most olympiad relationships are power laws, y = kx^n. A power law becomes a straight line on log–log axes, and the slope is the exponent:

$$\log y = \log k + n\log x. \tag{1}$$

This is why a log–log graph is a measuring instrument, not decoration. If a student collects planet data and plots log T against log a, the slope comes out as 1.5 and they have measured Kepler's third law.

Scaling also answers questions that look impossible. A sphere of radius R has area ∝ R² and volume ∝ R³. Stretch everything by a factor s and:

- area grows by s²,
- volume and mass grow by s³,
- the ratio {area}/{mass} ∝ s^(-1) falls.

That single ratio explains why insects cannot be the size of cars: the surface that supplies oxygen grows more slowly than the body that consumes it.

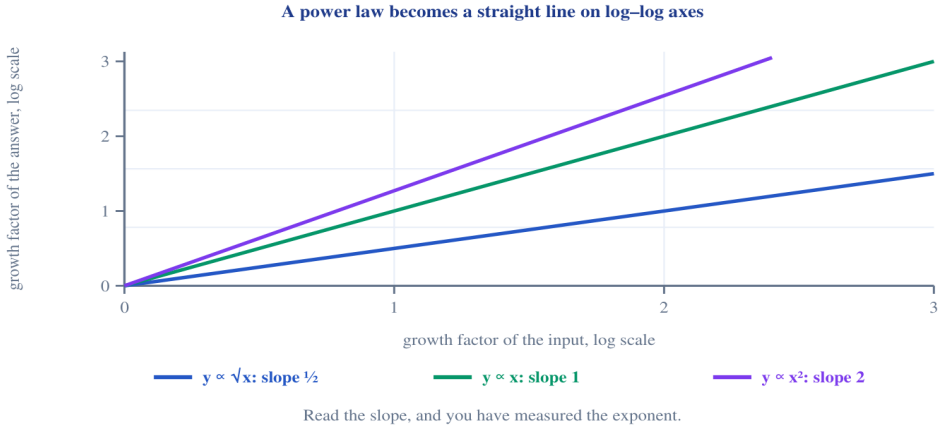


Figure 2. Doubling the radius of a sphere multiplies its area by 4 and its volume by 8. Reading the slope of a log–log graph tells you the exponent without any algebra.

4. Approximation you can defend

An approximation is only legitimate when you can say how large the error is. Three tools are enough for most of the syllabus.

Small parameter. If a quantity x is small compared with 1, then

$$(1+x)^{\alpha} \approx 1 + \alpha x,\ \sin x \approx x,\ e^{x} \approx 1 + x,\ \ln(1+x) \approx x. \tag{2}$$

To first order in x these are usually all you need, and they are enough to keep the leading correction — which is exactly what earns marks in estimation problems.

Taylor series. Around a point a,

$$f(a+\delta) \approx f(a) + f'(a)\delta + \frac{1}{2} f''(a)\delta^{2}. \tag{3}$$

The first term is the value, the second tells you the slope, the third tells you whether a stationary point is a minimum. This is also how you show that small oscillations are harmonic: near a minimum, $U(q) \approx U_0 + \frac{1}{2} U''(q_0)(q-q_0)^2$, and a quadratic potential means a spring.

Dimensional and physical limits. Push a parameter to 0 or ∞ and check the answer behaves as the physical situation does. If a formula for the speed of a rolling ball does not reproduce $v = \sqrt{2gh}$ as the radius goes to zero, something has been lost.

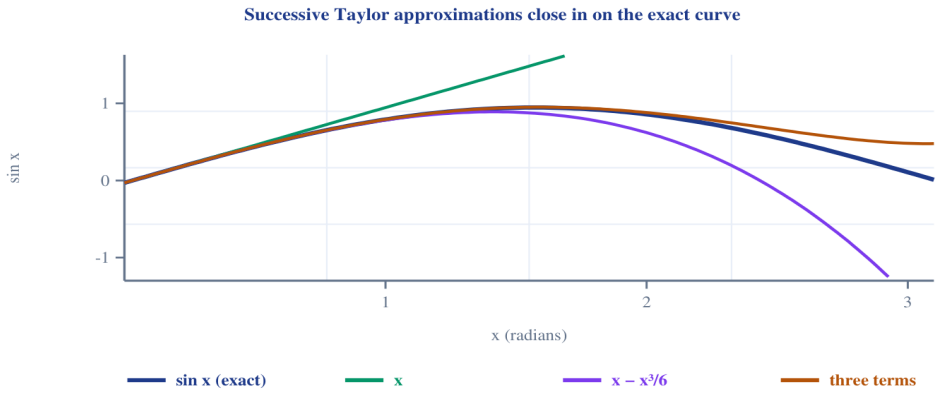


Figure 3. Each extra term buys accuracy over a wider range. Estimate the error before you decide whether the approximation is good enough for the precision you need.

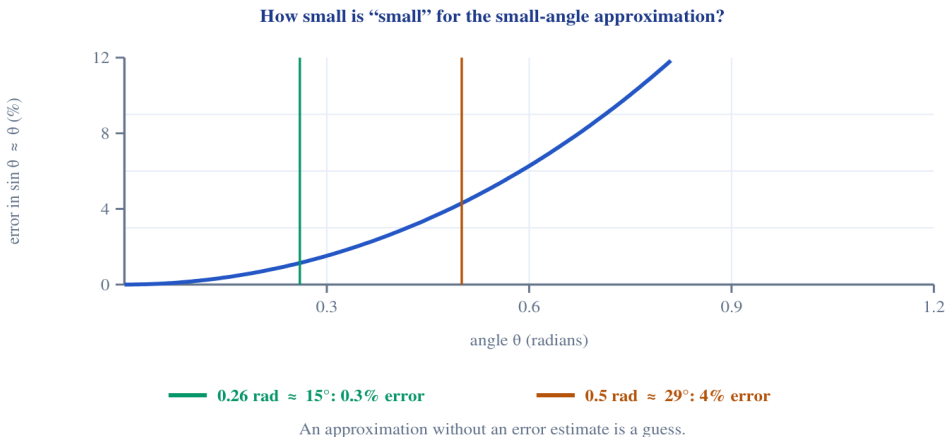


Figure 4. The error is below 1% only for angles under about 0.4 rad (23°). Quote that number instead of claiming the angle is “small enough”.

5. Order-of-magnitude estimates

Olympiad problems often ask for an estimate rather than a measurement. The method is a chain of ratios, each of which you know to within a factor of two or three:

$$\{\text{answer}\} = (\{\text{rate or density}\}) \times (\{\text{geometry}\}) \times (\{\text{correction}\}). \tag{4}$$

Two habits make estimates reliable. First, keep units visible so that the wrong numerators cancel audibly. Second, quote the uncertainty of the result — with four factors each uncertain by a factor of two, you cannot quote three significant figures honestly.

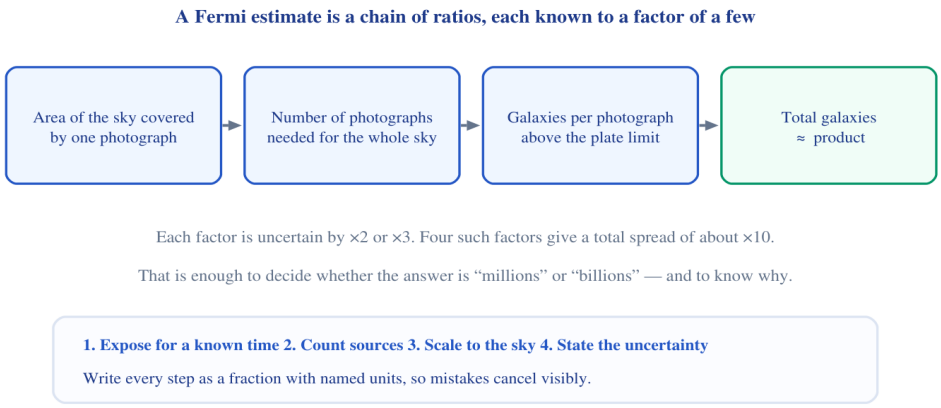


Figure 5. The point of a Fermi estimate is not the third digit; it is a defensible power of ten plus an honest uncertainty.

6. Symmetry and the method of choosing the right frame

If a configuration looks identical after a transformation, the physical result cannot distinguish the two arrangements, so the answer must be invariant under that transformation. This is worth more than an equation.

Three applications recur:

- **Mirror or rotational symmetry** removes a force component or a field component without solving anything.
- **Invariance in a free-fall frame** says a freely falling lift is equivalent to gravity-free space: spring, pendulum and scale questions often collapse.
- **Gauge-like shifts in potential** remind you that only potential differences do work.

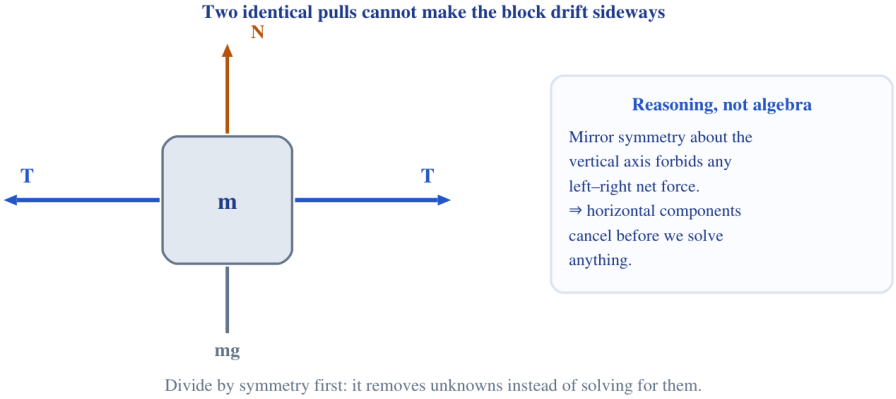


Figure 6. Olympiad problems are often built so that one symmetry kills most of the algebra. Look for what the configuration cannot distinguish.

The habit that separates strong solutions. Strong solutions carry a sentence for every step: **why** this model, **why** this approximation, **why** the discarded term is small. Mark schemes reward those sentences, and so does every real problem. A number without a reason is worth little; a reason with an approximate number is worth a lot.

7. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Dimensional fingerprint. The period T of a small oscillation of a mass m hanging on a spring of stiffness k depends on m and k. Newton's second law gives [k] = [M T^(-2)]. Use dimensional analysis to find how T depends on m and k, and state what the method cannot give you.

Worked solution →

[1] Problem 2 — How big is a star? A star has luminosity L and surface temperature T. The Stefan–Boltzmann law says the power radiated per unit area is σ T^4. Use dimensional reasoning on the relation L ∝ R²σ T^4 to check it, then explain why doubling T multiplies L by a larger factor than doubling R.

Worked solution →

[2] Problem 3 — The mass of a raindrop's argument. A pendant drop hangs from a tap of radius r. Surface tension γ (energy per unit area, with dimensions [M T^(-2)]) supports its weight. (a) Find the combination of γ, r and the gravitational acceleration g that has the dimensions of a mass. (b) Estimate the largest drop radius that can hang if γ is about 0.07 {N,m^(-1)} and a drop of radius 1 {mm} has mass 4×10^(-6) {kg}.

Worked solution →

[2] Problem 4 — Small-angle error budget.

Use **(a)** the first two terms of the Taylor series for sin θ to estimate the fractional error of the small-angle approximation at θ = 0.30 {rad}. (b) At what angle does the small-angle approximation overestimate sin θ by 1%? Use the next term to estimate the answer, then check it numerically.

Worked solution →

[2] Problem 5 — Reading a log–log slope. Two quantities measured in an experiment give the points (log x, log y) = (0, 1.30) and (2, 2.90) on a log–log graph (base 10, y in metres; x in seconds). (a) Find the exponent n of the power law y = kx^n. (b) Find k with its units. (c) Explain why a single wrong data point may barely affect this method but destroys a naive two-point calculation.

Worked solution →

[2] Problem 6 — Symmetry in a lift. A pendulum of length L, a block resting on a spring, and a bathroom scale are carried in a lift. The lift cable snaps and the car falls freely. Describe what the three items do, using the symmetry argument of a freely falling frame. Then explain why the conclusion would fail if the lift also had an acceleration of 0.5g downward.

Worked solution →

[2] Problem 7 — Fermi: number of stars in the observable sky. Estimate the number of stars a telescope could photograph. Use these generous guide numbers: the sky contains about 4×10^4 square degrees; a wide-field photograph covers 10 square degrees; a typical photograph shows 10^4 stars above its limit. (a) Estimate the total. (b) A real survey sees 10^9 stars. Explain the factor of order 10 discrepancy without doing any extra arithmetic.

Worked solution →

[2] Problem 8 — Power-law scaling of a body. A spherical animal is scaled up isotropically by a factor s. Heat is produced in proportion to volume but lost through the surface. (a) Find how the heat lost per unit mass depends on s. (b) Show that for a simple model with constant temperature, the time τ a body can keep its temperature depends on s — derive the exponent. (c) State one biological advantage and one disadvantage of being large in this model, and one reason the model is crude.

Worked solution →

[2] Problem 9 — The limits test that catches wrong formulas. A projectile is fired upward with speed v on a spherical body of radius R and mass M. Three proposed expressions for the maximum height h above the surface are

(i) $h = v^2R/(2GM - v^2R)$,

(ii) $h = vR/\sqrt{(GM)}$, (iii) $h = v^2/(2g)$ with $g = GM/R^2$ constant.

(a) Check dimensions of all three. (b) Test the limit $v \to 0$. (c) Test the limit $v^2R \to 2GM$ and explain which formula survives, and why.

Worked solution →

[3] Problem 10 — Building a model from scratch. A student claims that the radius of a crater made by an impactor should depend only on the impactor's kinetic energy E, the target density ρ and gravity g. (a) Use dimensional analysis to find the only possible combination. (b) Real craters in a deep ocean of water behave differently from craters in rock. Suggest one further quantity that must enter and explain what changes in your formula. (c) The student's formula has no dependence on the impactor's own radius. Explain why that can be true for a hypervelocity impact and why it must fail for a slow, gentle impact.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

Fermi problems (order-of-magnitude practice)

Background and examples for the estimation habits used in 01.01.